

The hole top mode

$v_{\text{top}} \propto 1/(\sqrt{\omega} P'_Y)$ is the top hole OP; it is not the excess of $\mathfrak{T}_0$

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Abstract

After r407–r411 the remaining P1-shaped statement is “at most one negative direction”. The candidate for that direction is the top orthogonal polynomial of the hole measure $(Y, u^\vee)$: $v_{\text{top},i} \propto 1/(\sqrt{u^\vee(y_i)} P'_Y(y_i))$, source-pure from Y , $u^\vee$, P'_Y (no eigenvector, no SVD, no target). The formula is a theorem over $\mathbb{Q}$. The Hole-Top-Mode test $Q_{\text{top}}(I - \mathfrak{T}_0^* \mathfrak{T}_0) Q_{\text{top}} \succeq 0$ **fails on flagship MAIN** ($\lambda_{\min} = -0.0476$, leftover $\sigma = 1.024$ on $w9$). Quadrature dominance already fails for $p \equiv 1$. One clean MAIN violation closes the route. No RH claim.

Status. Lemmas 1–2 are proved over $\mathbb{Q}$. Proposition 1 is linear algebra (the implication is not in doubt; its hypothesis is false). HTM on MAIN, QD, and “dead χ hold” are **refuted**. Ausgang TOP_MODE_REFUTED. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The formula

Let $Y = \{y_1, \dots, y_q\}$ be simple, $\omega_i > 0$, $P_Y(x) = \prod_k (x - y_k)$.

Lemma 1 (Lagrange identity). *For every polynomial p of degree at most $q - 2$,*

$$\sum_{i=1}^q \frac{p(y_i)}{P'_Y(y_i)} = 0.$$

On four rational atoms this is exact in $\mathbb{Q}$ for the monomial, Chebyshev, and Newton bases.

The residue of p/P_Y at infinity vanishes in that degree range, and the partial-fraction coefficient of $1/(x - y_i)$ is $p(y_i)/P'_Y(y_i)$.

Lemma 2 (Top hole OP). *The samples $\pi_{q-1}^Y(y_i) = c/(\omega_i P'_Y(y_i))$ are orthogonal to every polynomial of degree $\leq q - 2$ in the discrete inner product $\langle f, g \rangle_\omega = \sum_i \omega_i f(y_i) g(y_i)$. Gram–Schmidt in three bases yields the same samples up to scale. In the $\sqrt{\omega}$ -coordinate,*

$$v_{\text{top},i} \propto \frac{1}{\sqrt{\omega_i} P'_Y(y_i)}.$$

The constructor uses only Y and $u^\vee = \omega$. No eigendecomposition of C , no SVD of $\mathfrak{T}_0$, no target matrix.

2 The primary test

$\mathfrak{T}_0$ is the min-norm interpolant $Y \rightarrow X$ in $\mathcal{P}_{<d_0}$ (r409). Write $Q = I - vv^T$ for unit $v = v_{\text{top}}$.

Proposition 1 (Min-max). *If $\|\mathfrak{T}_0 z\|_X \leq \|z\|_Y$ for all $z \perp v_{\text{top}}$, then $\mathfrak{T}_0$ has at most one singular value > 1 . By the r407/r411 dictionary $\mathfrak{T}_0^* \mathfrak{T}_0 = C^{-1}$ this is $\lambda_2(C) \geq 1$, which is P1.*

The hypothesis is the projected Loewner condition $Q(I - \mathfrak{T}_0^* \mathfrak{T}_0) Q \succeq 0$, equivalently $\#\{\sigma(\mathfrak{T}_0|_{v^\perp}) > 1\} = 0$.

Flagship MAIN $w9$. $\text{corr}(v_{\text{top}}, C\text{-ev}_0) = 0.6766$, better than the r410 Fourier face (0.386 vs $C\text{-ev}_0$, 0.278 vs v_{top}). Projecting v_{top} cuts $\|\mathfrak{T}_0\| = 1.08014$ to $\sigma_{\text{max}}(v^\perp) = 1.02354$, still > 1 : $\lambda_{\text{min}} = -0.04763$, $n_{>1} = 1$. The Nyquist face is real and is not the excess (r411: the excess *is* $C\text{-ev}_0$).

Census. Core-42: P1 windows hold 4/28, PD 14/14 (trivial: $\|\mathfrak{T}_0\| \leq 1$ already). EXT P1 {119, 35, 109, 69} all break. Living χ : P1 1/37, PD 41/41. Dead χ hold 4/6; the two dead-P1 windows $\chi_3\text{-}19$ and $\chi_3\text{-}39$ break. Death is not an HTM-edge.

3 Quadrature dominance

Every $z \perp v_{\text{top}}$ is the $\sqrt{\omega}$ -sample of a unique $p \in \mathcal{P}_{\leq q-2}$. QD asks $\sum_X u^\vee p^2 \leq \sum_Y u^\vee p^2$. Min-norm gives $\|\mathfrak{T}_0 z\| \leq \|p\|_X$, so QD implies HTM.

On $w9$, already $p \equiv 1$ fails: $\sum_X u^\vee / \sum_Y u^\vee = 1.35713$. The min-norm gap *does* save this direction ($\|\mathfrak{T}_0 v_1\| = 0.359 < 1$) and does *not* save the leftover excess ($\sigma_{\text{max}}(v^\perp) > 1$). QD is strictly stronger than HTM, and both fail.

4 Kills

Permutation (same Y) and scramble break the primary test with $n_{>1} = 20$ and $\lambda_{\text{min}} \in \{-54, -217\}$. Dropping the signs of P'_Y leaves $\sigma_{\text{max}} \approx \|\mathfrak{T}_0\|$ (the Nyquist face dies). A random unit vector does the same. Substituting raw u for $u^\vee$ drops the $C\text{-ev}_0$ correlation to 0.126 and leaves $n_{>1} = 1$.

5 What this closes

The formula is a theorem, and v_{top} is the canonical top hole mode. It is not the one negative direction of P1. The reviewer order applies: one clean MAIN violation closes the hole-top-mode route. P1 remains $\iff \lambda_2(C) \geq 1$; that RHS is not this vector. No RH claim.